```
#include <stdio.h>
#include <stdlib.h>

typedef struct node {
   int data;
   struct node *left, *right;
} node;

node* root = NULL;

node* create(int val) {
   node* temp = (node*)malloc(sizeof(node));
   temp->data = val;
   temp->left = temp->right = NULL;
   return temp;
}

node* insert(node* root, int val) {
   if (root == NULL)
      return create(val);
   if (val < root->data)
      root->left = insert(root->left, val);
   else
      root->right = insert(root->right, val);
   return root;
}

node* findMin(node* root) {
   while (root->left != NULL)
      root = root->left;
   return root;
```

```
}

node* deleteNode(node* root, int val) {
   if (root == NULL)
      return NULL;
   if (val < root->data)
      root->left = deleteNode(root->left, val);
   else if (val > root->data)
      root->right = deleteNode(root->right, val);
   else {
      if (root->left == NULL)
         return root->right;
      else if (root->right == NULL)
         return root->left;
      node* temp = findMin(root->right);
      root->data = temp->data;
      root->right = deleteNode(root->right, temp->data);
   }
   return root;
}

void inorder(node* root) {
   if (root != NULL) {
      inorder(root->left);
      printf("%d ", root->data);
      inorder(root->right);
   }
}

void preorder(node* root) {
   if (root != NULL) {
```

```
        printf("%d ", root->data);
        preorder(root->left);
        preorder(root->right);
    }
}

void postorder(node* root) {
    if (root != NULL) {
        postorder(root->left);
        postorder(root->right);
        printf("%d ", root->data);
    }
}

void main() {
    int choice, val;

    while (1) {
        printf("\n--- BST MENU ---\n");
        printf("1. Insert\n2. Delete\n3. Inorder\n4. Preorder\n5. Postorder\n6. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);
        switch(choice) {
            case 1:
                printf("Enter value: ");
                scanf("%d", &val);
                root = insert(root, val);
                break;
            case 2:
                printf("Enter value to delete: ");
                scanf("%d", &val);
```

```
        root = deleteNode(root, val);
        break;
    case 3:
        printf("Inorder: ");
        inorder(root);
        break;
    case 4:
        printf("Preorder: ");
        preorder(root);
        break;
    case 5:
        printf("Postorder: ");
        postorder(root);
        break;
    case 6:
        exit(0);
    default:
        printf("Invalid choice!");
  }
}
```

```
--- BST MENU ---
1. Insert
2. Delete
3. Inorder
4. Preorder
5. Postorder
6. Exit
Enter choice: 1
Enter value: 50

--- BST MENU ---
1. Insert
2. Delete
3. Inorder
4. Preorder
5. Postorder
6. Exit
Enter choice: 1
Enter value: 30

--- BST MENU ---
1. Insert
2. Delete
3. Inorder
4. Preorder
5. Postorder
6. Exit
Enter choice: 1
Enter value: 70

--- BST MENU ---
1. Insert
2. Delete
3. Inorder
4. Preorder
5. Postorder
6. Exit
Enter choice: 1
Enter value: 20

--- BST MENU ---
1. Insert
2. Delete
3. Inorder
4. Preorder
5. Postorder
6. Exit
Enter choice: 1
Enter value: 40
```

```
--- BST MENU ---
1. Insert
2. Delete
3. Inorder
4. Preorder
5. Postorder
6. Exit
Enter choice: 3
Inorder: 20 30 40 50 70
--- BST MENU ---
1. Insert
2. Delete
3. Inorder
4. Preorder
5. Postorder
6. Exit
Enter choice: 2
Enter value to delete: 30

--- BST MENU ---
1. Insert
2. Delete
3. Inorder
4. Preorder
5. Postorder
6. Exit
Enter choice: 3
Inorder: 20 40 50 70
--- BST MENU ---
1. Insert
2. Delete
3. Inorder
4. Preorder
5. Postorder
6. Exit
Enter choice: 4
Preorder: 50 40 20 70
--- BST MENU ---
1. Insert
2. Delete
3. Inorder
4. Preorder
5. Postorder
6. Exit
Enter choice: 5
Postorder: 20 40 70 50
```

}